11.1.2 Sum of Digits using Recursion

A] Algorithm

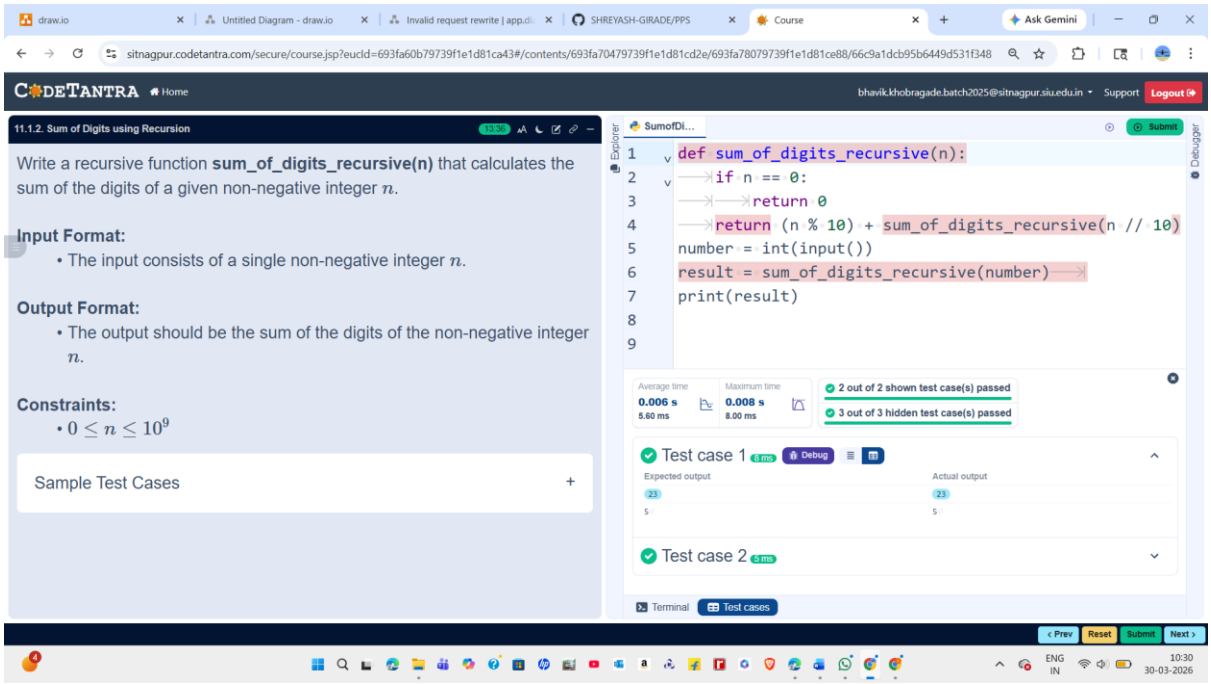
Step 1: Start
Step 2: Read an integer from the user and store it in number
Step 3: Define a recursive function sum_of_digits_recursive(n)
Step 4: Inside the function:
    Step 4a: If n == 0, return 0
    Step 4b: Otherwise, return (n % 10) + sum_of_digits_recursive(n // 10)
Step 5: Call sum_of_digits_recursive(number) and store the result in result
Step 6: Print result
Step 7: End

B**] code**

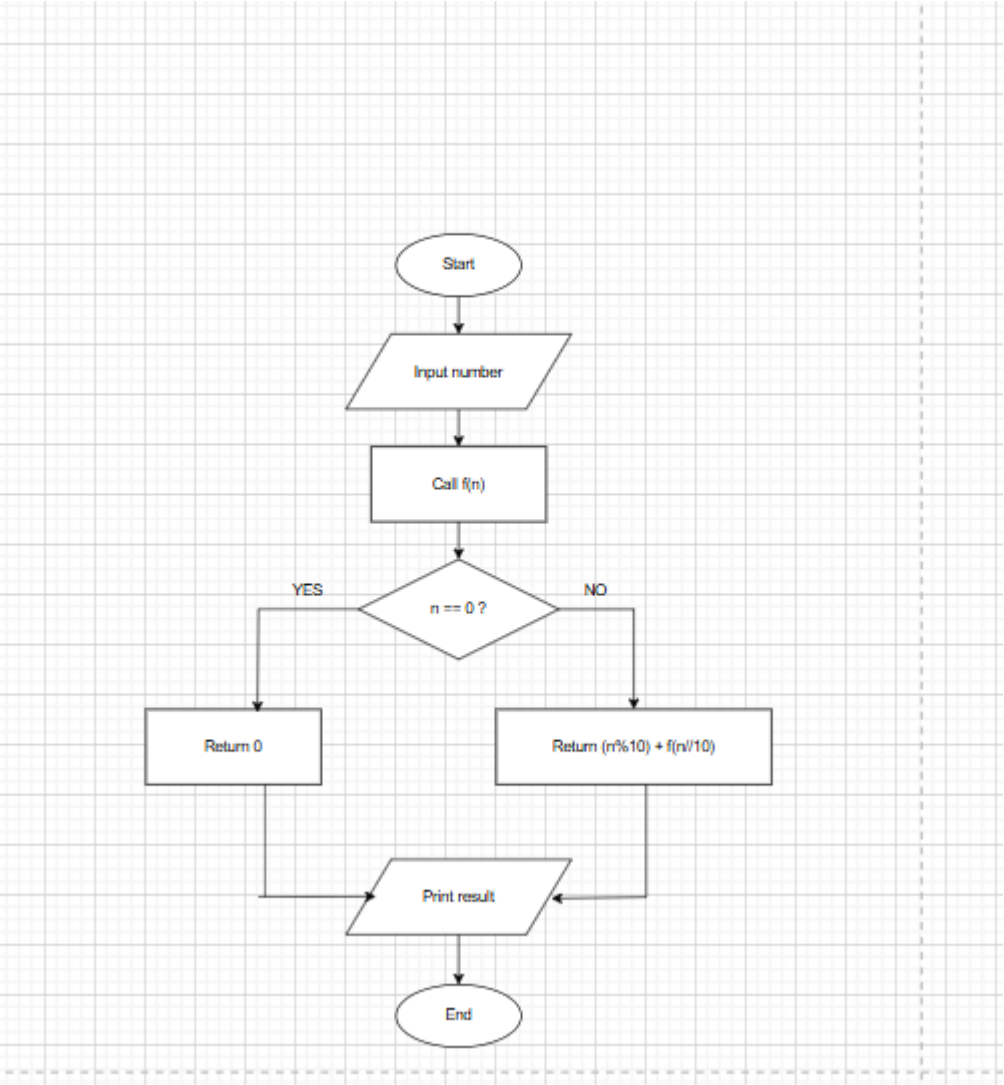
```
def sum_of_digits_recursive(n):
        if n == 0:
                return 0
        else:
                return  (n%10) +    sum_of_digits_recursive( n//10  )


number=int(input())
result = sum_of_digits_recursive(number)
print(result)
```

C] output



D] flowchart


Start
Input number
Call f(n)
n == 0 ?
YES
NO
Return 0
Return (n%10) + f(n/10)
Print result
End